Four criteria, four prices

All four decide the same question and any of them could serve as the definition. What separates them is cost — from not computable at all to cheaper than anything else on the page.

4

CRITERIA

TYPE	CONDITION	CONDITION	COST	USE IT WHEN
NOT A COMPUTATION				
 **The definition**	the statement itself	$\mathbf{x}^{\mathsf{T}} A\mathbf{x} > 0$ for every $\mathbf{x} \neq \mathbf{0}$	not directly testable	proofs and derivations
COMPUTABLE, EXPENSIVE				
 **Eigenvalue signs**	all $\lambda_i > 0$	every eigenvalue strictly positive	$O(n^3)$, iterative	the eigenvalues are wanted anyway
 **Leading principal minors**	every upper-left $k \times k$ determinant > 0	$\det A_k > 0$ for $k = 1, \ldots, n$	n determinants	small or symbolic matrices
COMPUTABLE, CHEAPEST				
 **Cholesky attempt**	the algorithm runs to completion	no negative under a square root	$\frac{1}{3}n^3$, terminating	whenever the answer is needed

They are equivalent, so the choice is purely practical: the definition for proving, the eigenvalues when they are wanted anyway, the minors for small symbolic matrices, and Cholesky whenever the answer is actually needed. That the cheapest test also produces a useful factorization is what makes it the standard one.